آمار و احتمال مهندسی

Engineering Probability & Statistics

Lecture Notes ▪ جزوه درسی

Module 9

Covariance and Correlation

کوواریانس و همبستگی

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Learning Objectives

After completing this module, students will be able to:

1. Compute and interpret covariance
2. Compute and interpret the correlation coefficient
3. Distinguish between independence and uncorrelatedness
4. Construct and interpret correlation matrices
5. Apply correlation analysis to engineering problems

9.1 Introduction: Measuring Linear Relationships

When two random variables are not independent, we need to quantify **how they relate**. Covariance and correlation measure the **linear relationship** between two RVs.

Engineering examples:

- Correlated noise in adjacent sensor channels
- Signal measurements at two antennas (MIMO correlation)
- Voltage and current in a circuit (related through impedance)
- Temperature and device performance (thermal dependence)

9.2 Covariance

Definition

$$\text{Cov}(X,Y) = E[(X-\mu_X)(Y-\mu_Y)] = E[XY] - E[X]E[Y]$$

Properties

1. $\text{Cov}(X,\, X) = \text{Var}(X)$
2. $\text{Cov}(X,\, Y) = \text{Cov}(Y,\, X)$ (symmetric)
3. $\text{Cov}(\text{aX} + b,\, \text{cY} + d) = \text{ac} \cdot \text{Cov}(X,\, Y)$
4. If $X,\, Y$ independent $\rightarrow \text{Cov}(X,\, Y) = 0$
5. $\text{Var}(X+Y) = \text{Var}(X) + \text{Var}(Y) + 2\,\text{Cov}(X,\, Y)$
6. $\text{Var}(X-Y) = \text{Var}(X) + \text{Var}(Y) - 2\,\text{Cov}(X,\, Y)$

Sign Interpretation

$\text{Cov}(X,\, Y)$	Meaning
> 0	X and Y tend to increase together
< 0	When X increases, Y tends to decrease
$= 0$	No linear relationship (uncorrelated)

Engineering Example: Amplifier Gain and Output

If gain G varies randomly and input is fixed at V_{in}:

- Output: $V_{\text{out}} = G \cdot V_{\text{in}}$
- $\text{Cov}(G,\, V_{\text{out}}) = \text{Cov}(G,\, G \cdot V_{\text{in}}) = V_{\text{in}} \cdot \text{Var}(G) > 0$

Higher gain $\rightarrow$ higher output (positive covariance).

9.3 Correlation Coefficient

Definition

$$\rho_{XY} = \frac{\text{Cov}(X, Y)}{\sigma_X \sigma_Y}$$

Properties

1. **Bounded:** $-1 \leq \rho \leq 1$
2. $\rho = 1$**:** Perfect positive linear relationship (Y = aX $+b$, $a > 0$)
3. $\rho = -1$**:** Perfect negative linear relationship (Y = aX $+b$, $a < 0$)
4. $\rho = 0$**:** Uncorrelated (no linear relationship)
5. **Dimensionless:** Unlike covariance, ρ doesn't depend on units

Interpretation Scale

	ρ
0.0 – 0.2	Very weak/negligible
0.2 – 0.4	Weak
0.4 – 0.6	Moderate
0.6 – 0.8	Strong
0.8 – 1.0	Very strong

Engineering Example: Correlated Fading

Two antenna paths with correlation $\rho = 0.3$:

- Low correlation $\rightarrow$ diversity works well (independent fading)
- High correlation $\rightarrow$ diversity gain is reduced

MIMO system design requires: $\rho < 0.5$ for effective spatial multiplexing.

9.4 Cross-Correlation

Definition

$$R_{XY} = E[XY]$$

Relationship to Covariance

$$\text{Cov}(X, Y) = R_{XY} - \mu_X \mu_Y$$

If either X or Y has zero mean: $\text{Cov}(X, Y) = E[XY] = R_{XY}$

Engineering Significance

Cross-correlation is fundamental in:

- Matched filters (correlation receivers in communications)
- Radar signal processing (detecting time delay)
- Signal detection (correlating received with known template)

9.5 Independence vs. Uncorrelatedness

The Critical Distinction

Statement	Always True?
Independent → Uncorrelated	✔ YES
Uncorrelated → Independent	✘ NO (in general)
Uncorrelated → Independent (jointly Gaussian)	✔ YES

Why Independent → Uncorrelated

If X, Y independent: $E[XY] = E[X]E[Y]$, so $\text{Cov}(X, Y) = E[XY] - E[X]E[Y] = 0$.

Counterexample: Uncorrelated but Dependent

Let $X \sim N(0, 1)$ and $Y = X^2$. Then:

- $\text{Cov}(X, Y) = E[XY] - E[X]E[Y] = E[X^3] - 0 = 0$ (since X is symmetric)
- But Y is completely determined by X → maximally dependent!

$\rho = 0$ **does NOT mean independent** (except for jointly Gaussian RVs).

The Gaussian Exception

For **jointly Gaussian** random variables:

$$\text{Uncorrelated} \Leftrightarrow \text{Independent}$$

This is why the Gaussian distribution is so convenient in engineering — checking $\rho = 0$ suffices to establish independence.

Engineering Implication

In communications with Gaussian noise:

- If noise samples are uncorrelated → they are independent
- This simplifies analysis enormously
- For non-Gaussian interference: uncorrelated does NOT guarantee independence

9.6 Correlation Matrix

Definition for Random Vector $X = [X_1, X_2, \ldots, X_n]^T$

The **covariance matrix** Σ:

$$\Sigma_{ij} = \text{Cov}(X_i, X_j)$$

$$\Sigma = \begin{bmatrix} \text{Var}(X_1) & \text{Cov}(X_1, X_2) & \cdots \\ \text{Cov}(X_2, X_1) & \text{Var}(X_2) & \cdots \\ \vdots & & \ddots \end{bmatrix}$$

The **correlation matrix** R (normalized):

$$R_{ij} = \rho_{X_i X_j} = \frac{\Sigma_{ij}}{\sqrt{\Sigma_{ii}\Sigma_{jj}}}$$

Properties of Covariance Matrix

1. Symmetric: $\Sigma = \Sigma^T$
2. Positive semi-definite: $x^T \Sigma x \geq 0$ for all x
3. Diagonal entries = variances
4. Off-diagonal entries = covariances

Engineering Example: Three Sensors

Three temperature sensors with:

- $\sigma_1 = \sigma_2 = \sigma_3 = 1°C$
- $\rho_{12} = 0.8$ (sensors 1,2 are nearby)
- $\rho_{13} = 0.3$ (sensor 3 is farther)
- $\rho_{23} = 0.4$

Correlation matrix:

$$R = \begin{bmatrix} 1.0 & 0.8 & 0.3 \\ 0.8 & 1.0 & 0.4 \\ 0.3 & 0.4 & 1.0 \end{bmatrix}$$

9.7 Variance of Linear Combinations

General Formula

$$\text{Var}\left(\sum_{i=1}^{n} a_i X_i\right) = \sum_{i=1}^{n} a_i^2 \text{Var}(X_i) + 2\sum_{i<j} a_i a_j \text{Cov}(X_i, X_j)$$

In matrix form: $\text{Var}(a^T X) = a^T \Sigma a$

Special Case: Sum of Two RVs

$\text{Var}(X+Y) = \text{Var}(X) + \text{Var}(Y) + 2\,\text{Cov}(X,\,Y) = \sigma_X^2 + \sigma_Y^2 + 2\rho\sigma_X\sigma_Y$

Correlation	$\text{Var}(X+Y)$	Effect
$\rho = 1$	$(\sigma_X + \sigma_Y)^2$	Maximum — variances add constructively
$\rho = 0$	$\sigma_X^2 + \sigma_Y^2$	Independent addition
$\rho = -1$	$(\sigma_X - \sigma_Y)^2$	Minimum — cancellation

Engineering Application: Averaging Correlated Sensors

If n sensors have equal variance σ^2 and pairwise correlation ρ:

$$\text{Var}(\bar{X}) = \frac{\sigma^2}{n}[1 + (n-1)\rho]$$

- If $\rho = 0$ (independent): $\text{Var}(\bar{X}) = \frac{\sigma^2}{n}$ (full averaging benefit)
- If $\rho = 1$ (identical): $\text{Var}(\bar{X}) = \sigma^2$ (no benefit from averaging!)
- If $\rho = 0.5$, $n = 4$: $\text{Var}(\bar{X}) = \frac{\sigma^2(1+3\times0.5)}{4} = 0.625\sigma^2$ (reduced benefit)

9.8 Practice Problems

Problem 1

X and Y have: $E[X] = 2$, $E[Y] = 3$, $\text{Var}(X) = 4$, $\text{Var}(Y) = 9$, $E[XY] = 8$.

(a) Find $\text{Cov}(X, Y)$. (b) Find ρ. (c) Find $\text{Var}(X + Y)$. (d) Find $\text{Var}(2X - 3Y)$.

Solution

(a) $\text{Cov}(X, Y) = E[XY] - E[X]E[Y] = 8 - 2(3) = 2$

(b) $\rho = \dfrac{2}{\sqrt{4} \cdot \sqrt{9}} = \dfrac{2}{6} = \dfrac{1}{3}$

(c) $\text{Var}(X + Y) = 4 + 9 + 2(2) = 17$

(d) $\text{Var}(2X - 3Y) = 4(4) + 9(9) + 2(2)(-3)(2) = 16 + 81 - 24 = 73$

Problem 2

Four identical sensors ($\sigma^2 = 1$) have pairwise correlation $\rho = 0.5$. What is $\text{Var}(\bar{X})$?

Solution $\text{Var}(\bar{X}) = \dfrac{\sigma^2}{n} \cdot [1 + (n-1)\rho] = \dfrac{1}{4} \cdot [1 + 3(0.5)] = 0.25 \times 2.5 = 0.625$

Compare: if independent ($\rho = 0$): $\text{Var}(\bar{X}) = 0.25$. Correlation reduces the effectiveness of averaging.

Problem 3

Given $X \sim N(0, 1)$, define $Y = X$ when $|X| < 1$ and $Y = -X$ when $|X| \geq 1$. Show that $\text{Cov}(X, Y) \neq 0$ in this case but that $E[X \cdot Y]$ can be computed.

Solution $E[XY] = E[X^2 \cdot 1(|X| < 1)] + E[-X^2 \cdot 1(|X| \geq 1)] = E[X^2 \cdot 1(|X| < 1)] - E[X^2 \cdot 1(|X| \geq 1)]$

$= P(|X| < 1)E[X^2 || X\ | < 1] - P(|X| \geq 1)E[X^2 || X\ | \geq 1]$

This is nonzero since $E[X^2 || X\ | < 1] \neq E[X^2 || X\ | \geq 1]$, so $\text{Cov}(X, Y) = E[XY] - 0 \neq 0$.

X and Y are clearly dependent (Y is defined from X), and in this case also correlated.

Next Module: Module 10 — Multiple Random Variables